

Element B

Identification & Analysis of Prior Solutions

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Problem Statement

Professionals in the medical field lack a fast and risk-free method for physical experience, essential for surgery training to be more efficient with patient care. Medicare currently pays \$180,000 per medical resident each year, a total of over 25 billion dollars,¹ and a great proportion of that is spent on the acquisition and usage of cadavers, a lengthy process with ethical and practical issues.

Introduction

Various previous virtual reality (VR) haptic feedback-focused and/or medical training-focused patents and products were analyzed based on their key functionalities, positives, and negatives. Based on their medical aptitudes, technological advancements in digital touch, and other categories, these were compared to determine the services provided in the current market. This evaluation of preexisting solutions allows greater comprehension of the task and what an effective solution would require.

Patents Covered

1. Exo-Tendon Haptic Glove Device
2. Mixed Reality Training System
3. VR Jamming Garment/Glove
4. Medical VR Surgical System
5. Pneumatic Augmented Reality Tactile Feedback Platform
6. Augmented Reality Patient Care System

Products Covered

1. bHaptics TactGlove DK1
2. HapticVR Platform
3. HaptX Gloves G1
4. Anatomage Table
5. Fluid Reality Haptic VR Glove
6. TouchDIVER Haptic Gloves

¹“The Costs of Training Internal Medicine Residents in the United States.” The American Journal of Medicine, U.S. National Library of Medicine, pubmed.ncbi.nlm.nih.gov/25063649/. Accessed 12 Nov. 2024.

I. Patent Research

Patent 1: Exo-Tendon Haptic Glove Device

Buchanan, T. F., & Meyer, T. (2019a, January 19). *Exo Tendon Motion Capture Glove Device with Haptic Grip Response*.

Patent Number: US 2018/0311570 A1

Patent Summary:

The design utilizes motors with strings attached to multiple spots on each finger. Each of these motors connects to a computer which calculates the position of each phalanx and creates the sensation of touching virtual objects via the retractions of the strings.

Patent Critique:

This design is focused on giving its user a large amount of feedback to the hands, within the fingers. Due to having a motor on each part of each finger, the hands are given a complete sense of what the virtual world entails. However, due to the large amount of motors necessary, the contraption is not contained to the hands and must extend onto the forearms. In addition to that the weight of the gloves limits the mobility of the user.

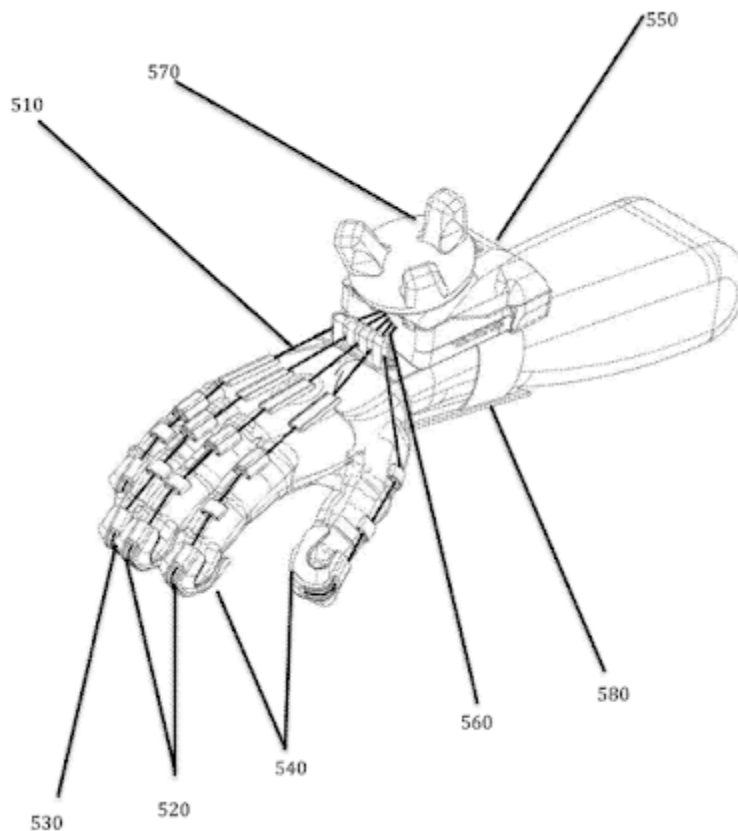


Figure 1.1: An illustration of the product showing its feedback system as well as the system that locates the device in 3D space.

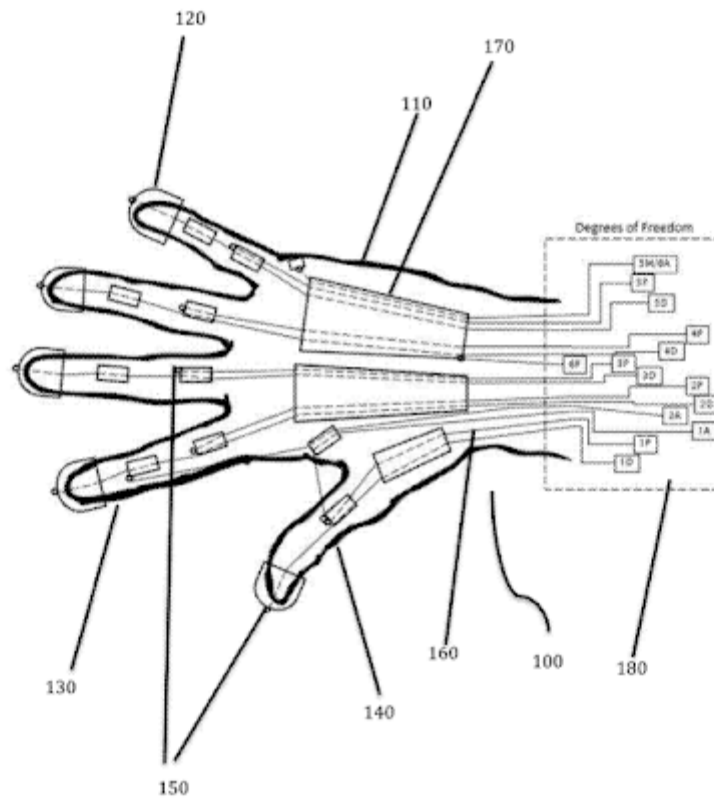


Figure 1.2: An illustration of the mounting locations of each string as well as where it will connect to each finger.

Patent 2: Mixed Reality Training System

Sela, G., Kerins, F., Abhari, K., & Srimohanarajah, K. (2018, April 26). *MIXED REALITY TRAINING SYSTEM*.

Patent Number: US 10,828,107 B2

Patent Summary:

The Mixed Reality Training System used a combination of physical objects as well as overlaid virtual projects to improve the realism of the training. It uses a tracking system to locate the trainee's position within space and combined with a computer, it calculates how the augmented objects should adjust. The physical model also can adapt due to the user's actions, such as parts of the model being detachable or being able to cut the model. The computer also analyzes the surgery to give the trainee a score based on their performance in the training simulation.

Patent Critique:

The grading system allows for a quicker turnaround time for trainees to know how they performed in their surgery. This would allow for quicker and more effective improvements over time. The physical model aids in realism but increases the complexity of setting the device up as each person requires a physical model. Also, because the physical model can be modified, parts of the model can't be used multiple times which increases the cost per usage.

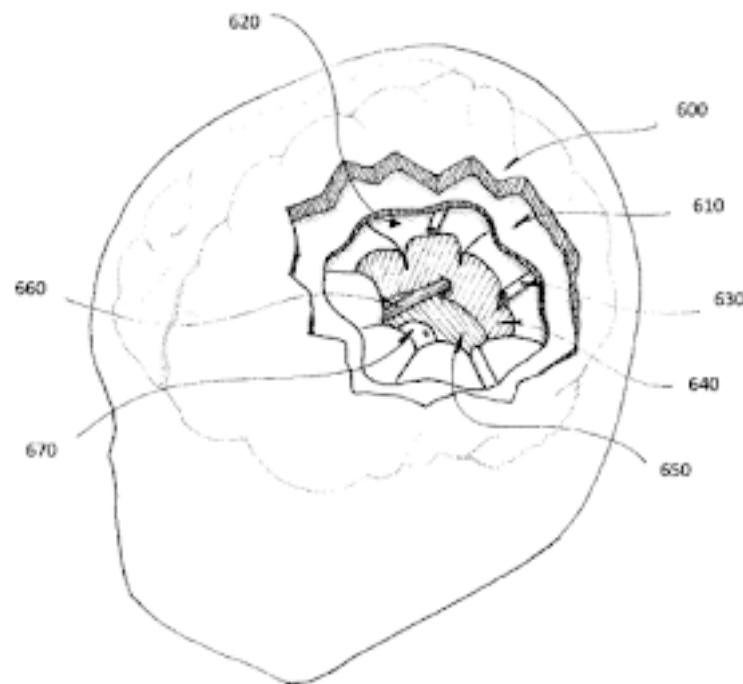


Figure 2.1: An illustration of the physical model used for a brain tumor extraction training system.

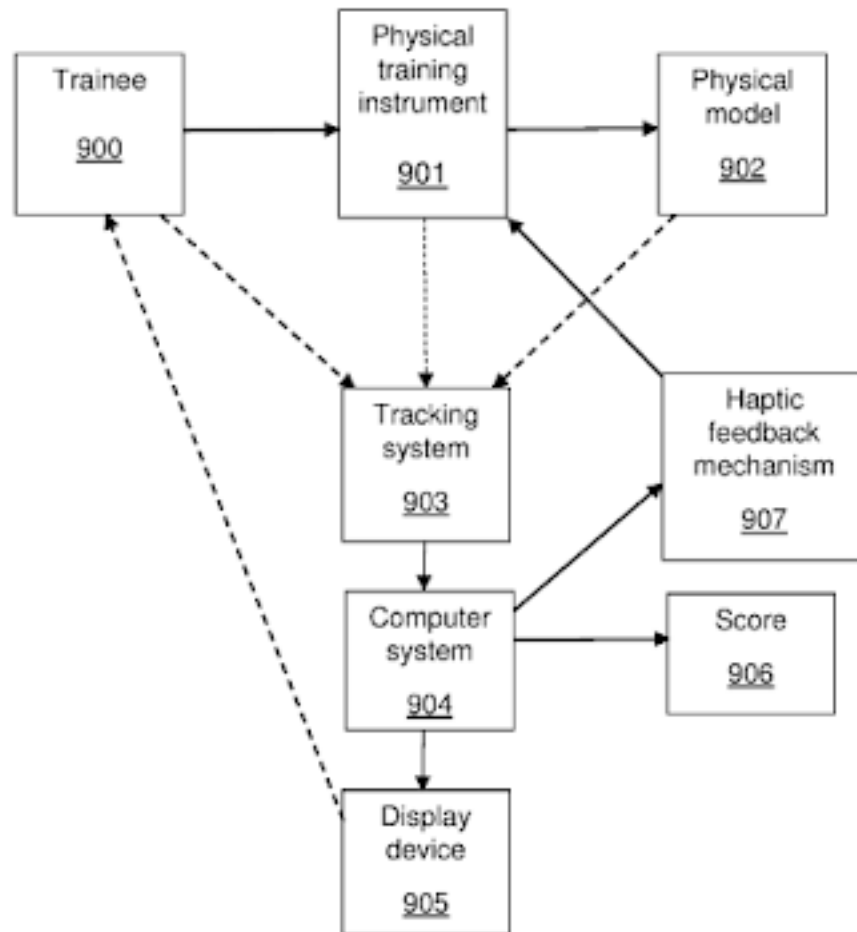


Figure 2.2: A flowchart illustrating the flow of information within the computer and mechanical system.

Patent 3: VR Jamming Garment/Glove

Keller et al. (2021, April 13). *Virtual Reality Garment Capable Of Jamming User Movement*.

Patent Number: US 10,976,826 B1

Patent Summary:

The patent outlines VR gloves that use various 'jamming mechanisms' to simulate touch in varied digital applications. Inflatable tendons (Figure 3.2), small microcup structures (Figure 3.4), and non-Newtonian fluid (Figure 3.5) held in high-jitter bags restrict movement to simulate the immobility and sensation of holding solid objects. Twisted strings (Figure 3.3) around tendons restrict movement further with added force: greater torque generates greater resistance, simulating varied rigidity. Anisotropic materials simulate force direction push back against over-resistance. Electromagnetic jamming mechanisms remove some reliance on electronic control, allowing precise adjustable resistance to hand movement.

Overall, each component simulates a different form of sense, collaborating to balance different forces of immobility, touch sensation, and texture. The glove itself has embedded sensors that monitor position, orientation, and motion, and can work wirelessly or through cables with a VR headset.

Patent Critique:

Pros:

- Adaptability for various forms of touch
- Able to work wirelessly
- Modular design allows for customization and upgrades
- Lightweight design for prolonged use
- Made by Facebook Technologies
 - Potential compatibility to previously released commercial products such as the Meta Quest VR headset series

Cons:

- Expensive parts used
 - High production costs would lead to a higher sales price than alternatives
- Battery life shortens on wireless usage
 - Surgical training would take hours of use, would restrict
- Wired usage restricts hand movement because of tugging
- Reliance on VR consoles removes the ability to adapt to AR advancements
- Potential overheating of parts
- Multiple parts work in tandem, allowing for a greater chance of error with any part completely messing up the system
 - All parts are powerful and pose serious harm if malfunctioned
 - No safety system

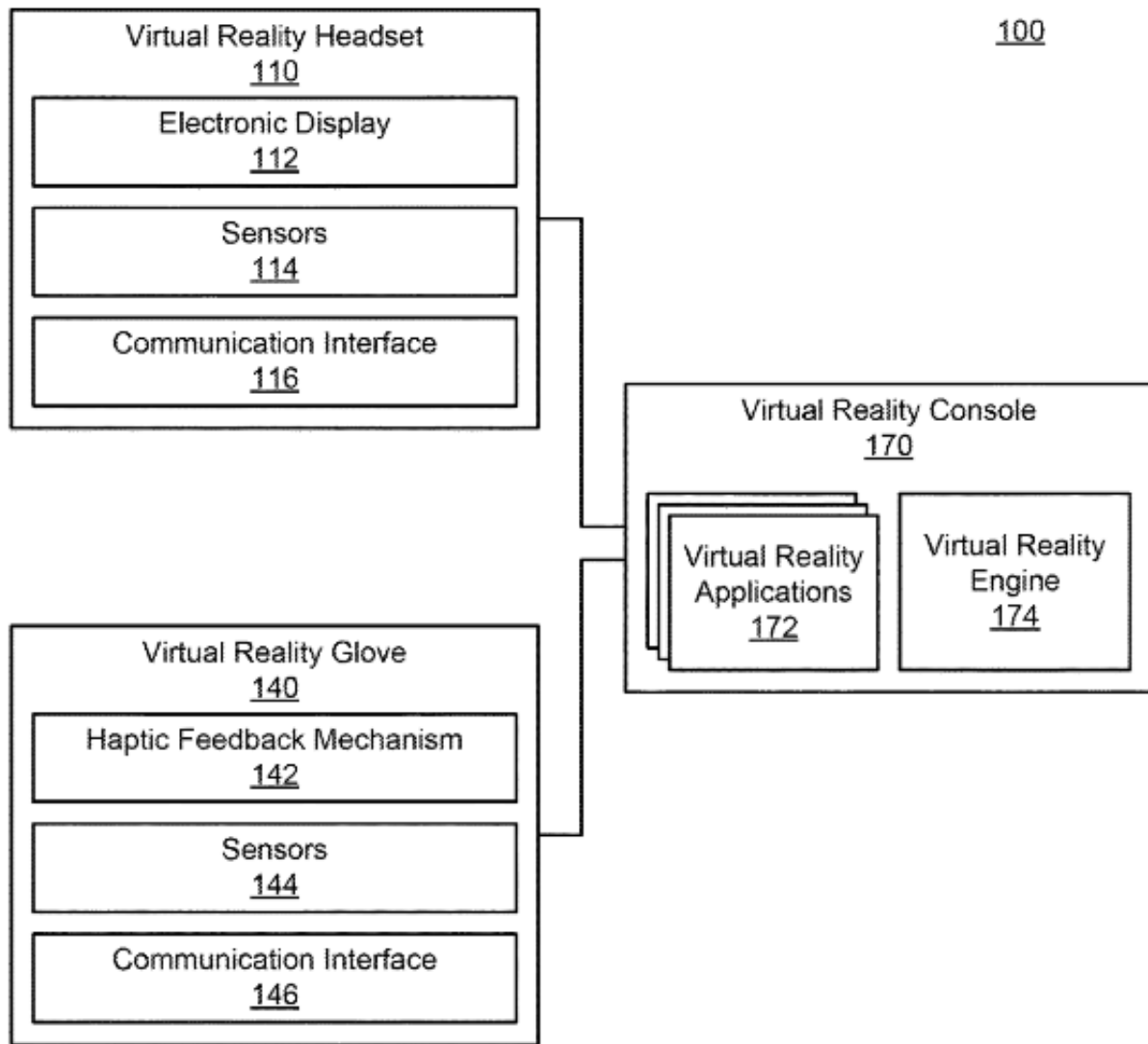


Figure 3.1: A block diagram describing each component of the virtual reality system.

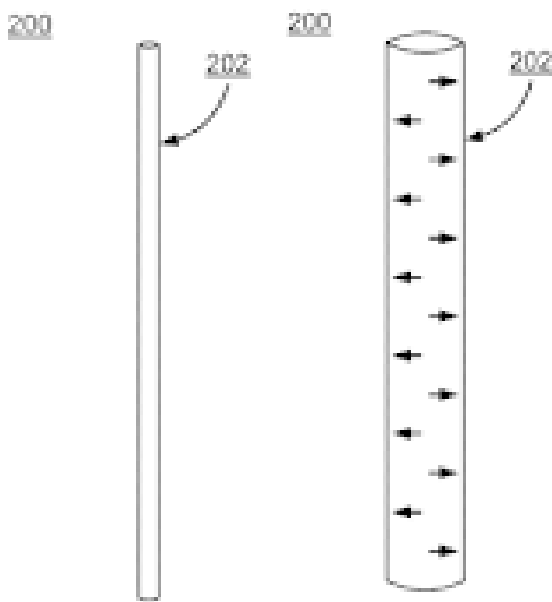


Figure 3.2: An illustration of an inflatable tendon. Left tendon is unjammed/uninflated. Right tendon is jammed and filled with air.

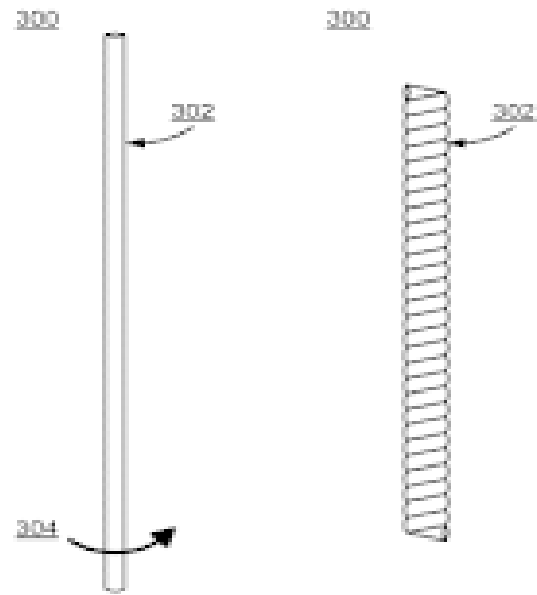


Figure 3.3: An illustration of a twisted string system. Left is unjammed/untwisted. Right is jammed, coiled around a tendon.

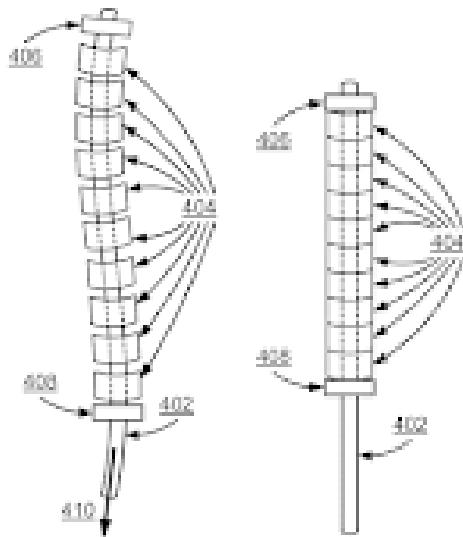


Figure 3.4: An illustration of a series of microcups. Left system is unjammed. Right system is jammed and interlocked together.

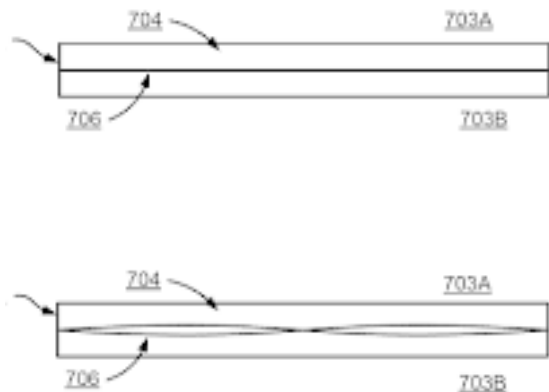


Figure 3.5: An illustration of a bag of non-Newtonian fluid. A filament (706) is immersed within the non-Newtonian fluid, and attached to an oscillator (not pictured). Top bag resting/unjammed. Bottom bag is oscillating, causing the fluid to solidify.

Patent 4: Medical VR Surgical System

Goel et al. (2022, May 12). *Medical VR Surgical System*.

Patent Number: US20220148448A1

Patent Summary:

This patent outlines a medical VR surgical system that simulates medical procedures based on patient-specific virtual anatomy: using medical imaging, the system can construct bones, tissues, and other anatomical structures. The system uses a headset, and controllers with haptic feedback that simulate the resistance of cutting/drilling.

The system can also provide metrics based on actions done in the simulation to evaluate success or failure. Cloud-based data storage is included, allowing collaboration as well for other experts to view one's work.

Patent Critique:

Pros:

- Realistic simulation
- Haptic feedback integration to mimic resistance
- Collaborative space
- Customizable
- Iterative learning functionality with feedback metrics
- Scalable to various medical fields
- Supports AR functionality

Cons:

- Expensive parts used
- Huge learning curve with the software
 - Not as close to reality; unintuitive
- Network and imaging dependencies
- Privacy concerns with cloud storage
- Haptic feedback isn't close to real touch
 - Controllers are not actually tracking finger motion
 - No real force on the hands

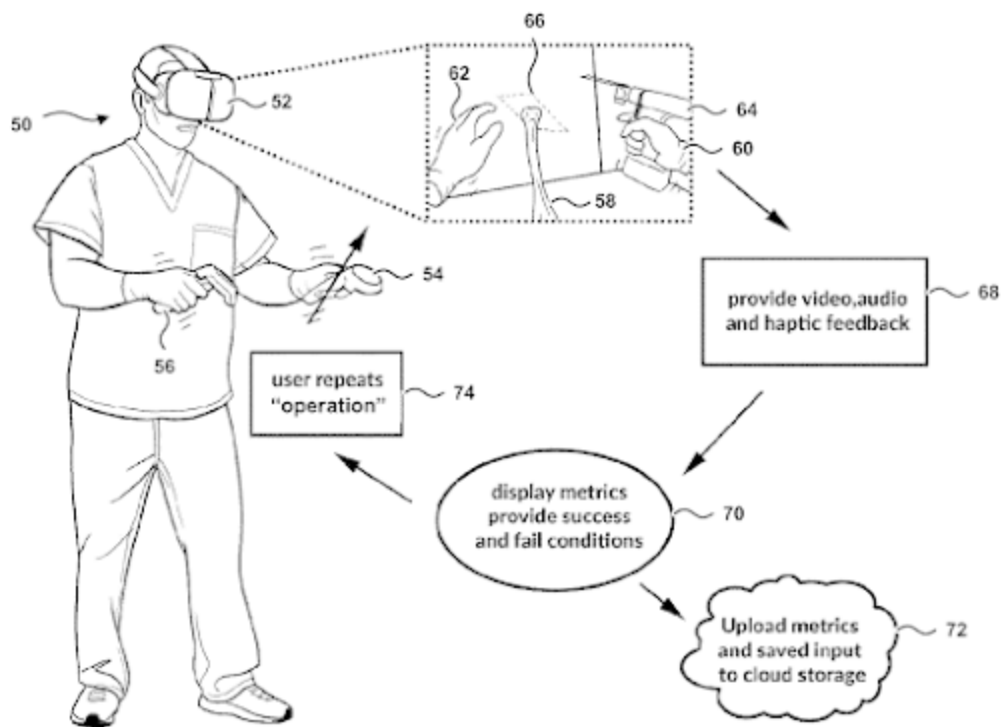


Figure 4.1: A diagram showcasing the process of a virtual surgical procedure.

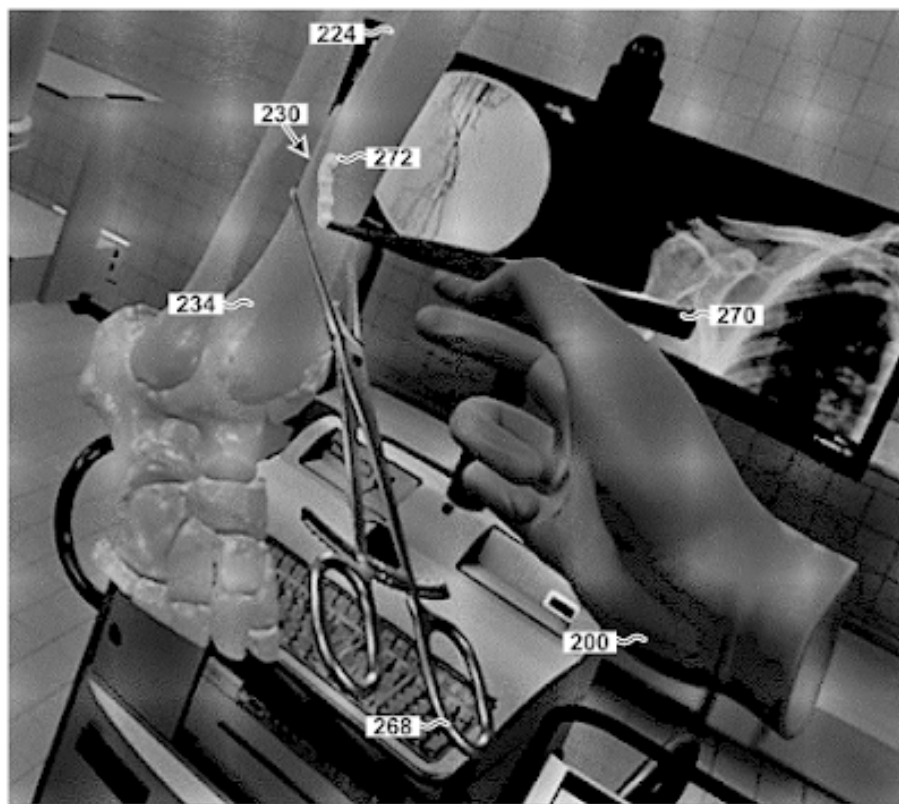


Figure 4.2: A digital image/screenshot from the simulation.

Patent 5: Pneumatic Augmented Reality Tactile Feedback Platform

Kerr et al. (2017, October 19). *Haptic glove using fluid-driven actuators for virtual reality and augmented reality applications*.

Patent Number: US20170300115A1

Patent Summary:

This device uses fluid dynamic pneumatics to exert pressure and provide tactile feedback. It uses pressure sensors for each finger and an actuating mechanism to control the speed at which the fluid (can be gas, air, or liquid) flows, allowing for high and low pressure for different objects. Additionally, the control panel can simulate external pressure changes and can support multiple compartments, allowing for a highly versatile and ergonomic system that can be worn as a glove, vest, helmet, or more.

Patent Critique:

Since this patent uses fluids to deliver pressure, it will need to be refilled more times than a simple mechanical glove with wires or electronics. It is also bulky and uses many expensive parts, such as solenoids, compressors, and small electronic components that could be easily damaged over use. However, this glove can exert precise pressure for different textures or hardness of objects due to its special “fluid” pressure. It also has fewer risks of overheating or short-circuiting, which could be useful in simulating wet or dry conditions. By combining this device with VR, it can properly simulate pressure, resistance, and texture variations corresponding to skin, tissue, or bone without practicing on one-time-use cadavers, while also providing real-time feedback.

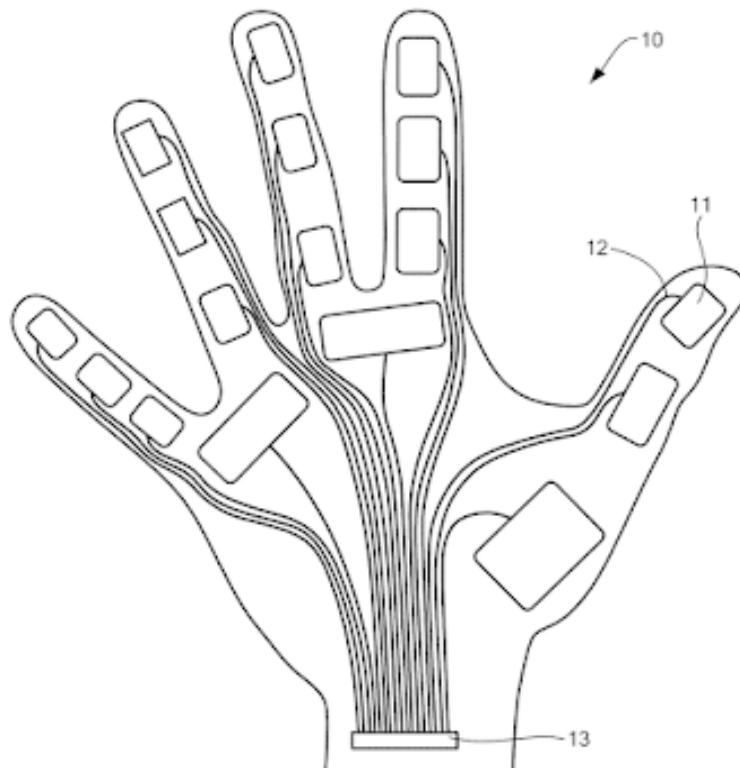


Figure 5.1: Illustration of the glove design with heat-sealed polyethylene pouches for each phalange and pneumatic cables for haptic feedback.

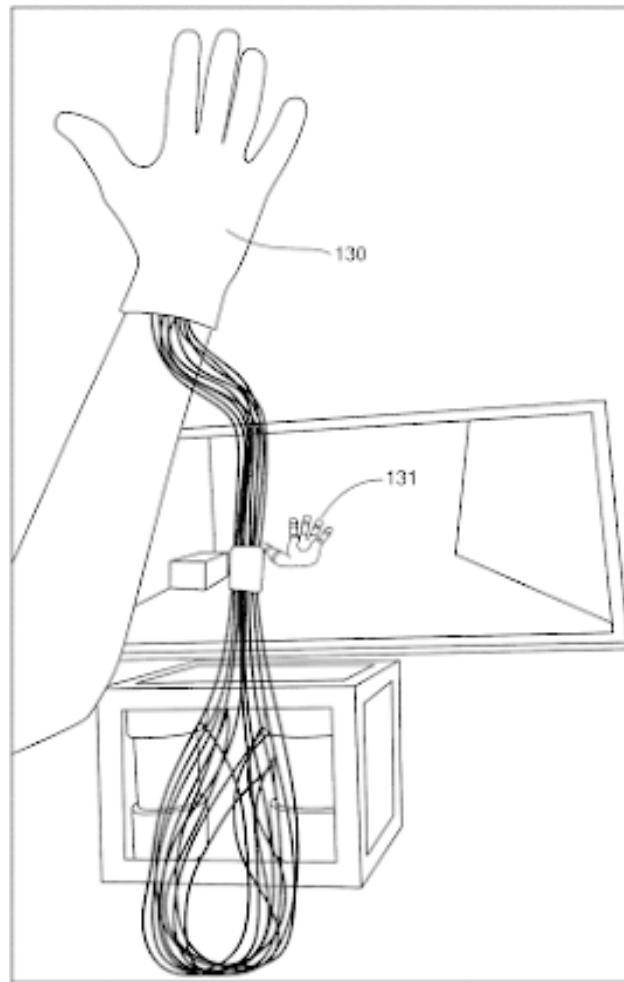


Figure 5.2: Illustration of a user's hand being tracked by a VR environment in Unity.

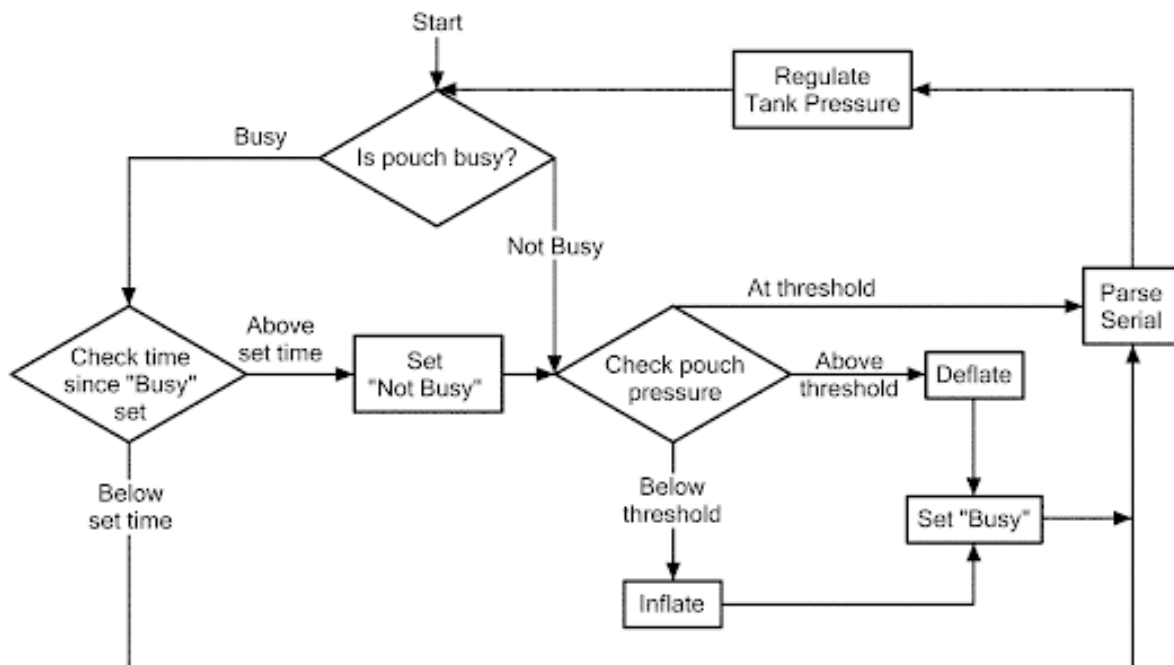


Figure 5.3: Diagram showing software logic used to regulate and adjust pouch inflation based on pressure readings from sensors.

Patent 6: Augmented Reality Patient Care System

Eggert et al. (2023, August 9). *Augmented reality system for teaching patient care.*

Patent Number: EP3669344B1

Patent Summary:

This AR Patient Care System integrates both physical and virtual anatomical models to create an interactive and realistic training environment. It overlays both models on top of each other, which allows the user to visualize and interact with the anatomy in real-time. By using tracking devices and optic feedback, the system creates coordinates and adjusts the virtual content, which can be sent to monitors outside the AR system for instructional review. While non-human interactive formats can help with teaching patient care, this patent simulates all the possibilities to teach students without the risks associated with real patients.

Patent Critique:

The augmented reality system boosts the training experience from the overlay, helping trainees better understand complex situations. It provides a safe learning environment with real-time feedback, as well as being cost-effective, reusable, and adaptable to any person. However, this system heavily relies on hardware and software, which can both be expensive and time-consuming to set up or fix, and both must be calibrated properly to ensure the virtual and physical models overlay correctly. Despite these shortcomings, each scenario and interaction can all be unique and can be videotaped for future reference, making this system a reliable device.

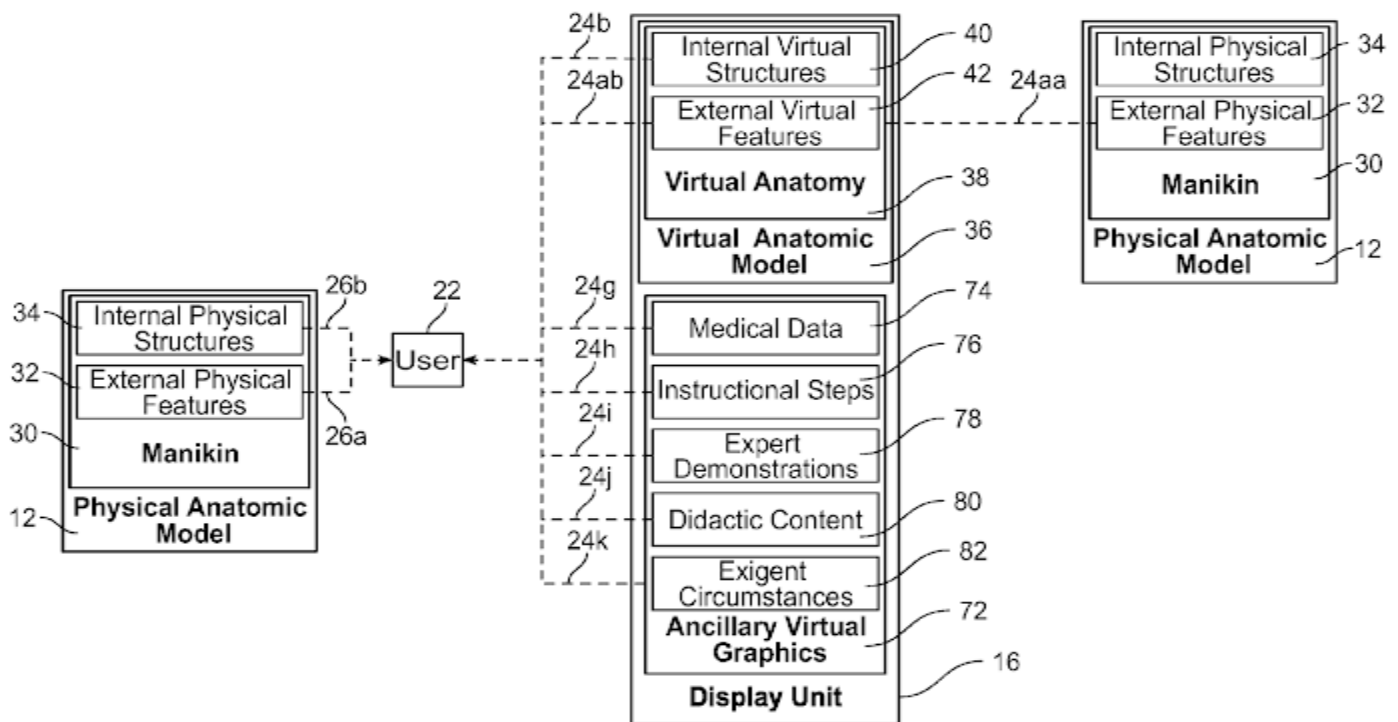


FIG. 2

Figure 6.1: Diagram illustrating a user receiving optic feedback from the virtual and physical anatomic models.

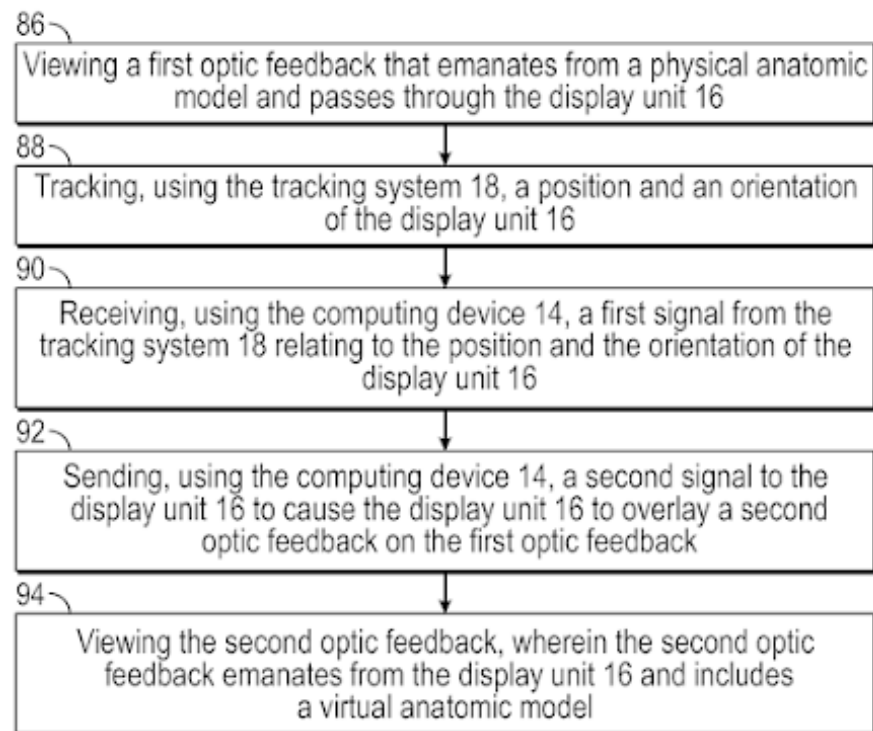


Figure 6.2: Sample flow diagram of the operating system.

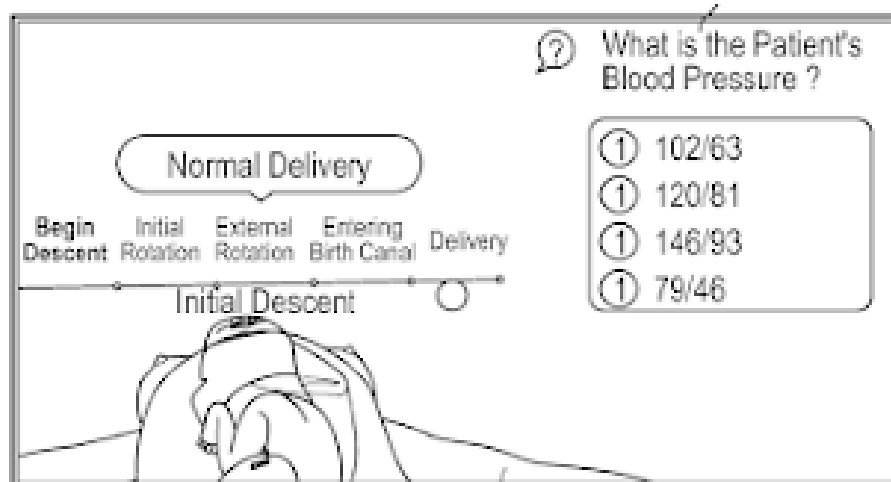


Figure 6.3a: Example question being asked in the “watch mode” of the AR system

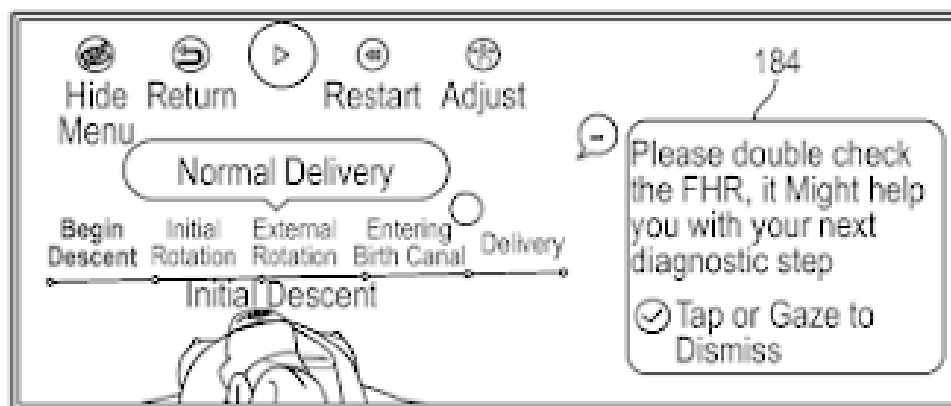


Figure 6.3b: Example message shown being delivered to the AR headset device

II. Product Research

Product 1: bHaptics TactGlove DK1

The TactGlove, made by bHaptics, is aimed at being a virtual reality glove that can bring a high level of feedback to users. It does this by creating vibrations at the tip of each finger and at the base of the hand to mimic the sensation of touching a surface. The glove has a sensor that can connect to a virtual reality headset with Bluetooth to a virtual reality headset and it is designed such that it can be tracked by built-in hand tracking to all-in-one headsets such as the Oculus Quest 3.^{2 3}

Specifications:

- 6 feedback points per hand
- 3.5 Hours of Battery Time
- Full Charge in 2 Hour
- 4 different hand sizes to support a variety of hand sizes.
- Removable Inner lining

Pros:

- Lightweight, 107g-115g
- Long battery life allows for long-use sessions
- Easy to put on and take off
- Removable inner layer for safe cleaning

Cons:

- Vibrations being the main force of feedback limits the extent of its practicality
- The price of 300 dollars limits who can purchase this device
- The removable inner layer creates an additional point of failure⁴

² Carter, R. (2024, January 12). *The Best VR Gloves in 2024: Harnessing Haptic Feedback*. XR Today. Retrieved December 6, 2024, from <https://www.xrtoday.com/virtual-reality/the-best-vr-gloves-in-2024-harnessing-haptic-feedback/>

³ bHaptic. (2024). *bHaptics TactGlove DK1 Wireless Haptic Gloves (Pair) - VR Accessory*. KnoxLabs. Retrieved 12 06, 24, from <https://www.knoxlabs.com/products/bhaptics-tactglove-dk1-haptic-gloves?srltid=AfmBOooZhpYmoK6P69NOkd7TPpjTSU4FyB2ZjaAdarD8ZvzBr73il8YI>

⁴ Sprigg, S. (2021, 12 29). *bHaptics unveils its TactGlove consumer-ready haptic gloves for VR*. Auganix. Retrieved 12 6, 2024, from <https://www.auganix.org/bhaptics-unveils-its-tactglove-consumer-ready-haptic-gloves-for-vr/>



Figure 7.1: An image of the gloves showing a view of both the front and back of the glove. As well as the housing of the electronics.⁵



Figure 7.2: An image of the different parts that compose the glove. The left picture shows the innermost layer. The middle picture shows the middle layer. The right picture shows the outer layer of the glove.⁶

⁵ bHaptic. (2024). *bHaptics TactGlove DK1 Wireless Haptic Gloves (Pair) - VR Accessory*. KnoxLabs. Retrieved 12 06, 24, from <https://www.knoxlabs.com/products/bhaptics-tactglove-dk1-haptic-gloves?srsId=AfmBOooZhpYmoK6P69NOkd7TPpjTSU4FyB2ZjaAdarD8ZvzBr73il8YI>

⁶ bHaptic. (2024). *bHaptics TactGlove DK1 Wireless Haptic Gloves (Pair) - VR Accessory*. KnoxLabs. Retrieved 12 06, 24, from <https://www.knoxlabs.com/products/bhaptics-tactglove-dk1-haptic-gloves?srsId=AfmBOooZhpYmoK6P69NOkd7TPpjTSU4FyB2ZjaAdarD8ZvzBr73il8YI>

Product 2: HapticVR

The Haptic VR platform is a product developed by Fundamental Surgery that can be integrated into Fundamental Surgery's existing product library, or be used alone. Utilizing two user interfaces mounted on a table, the HapticVR system can accurately mimic the tools used during surgery.⁷

Specifications:

- Two table-mounted feedback devices
- Able to be used with any off-the-shelf VR headset
- A part of the FundamentalVR ecosystem
- Premade user interfaces mimicking surgical tools such as scalpels
- Able to replicate soft tissue sensations

Pros:

- Usable with other FundamentalVR products
- Corded Solution so no limited battery life
- Wide range of connections
- Proven to improve training of medical staff
- Force feedback is not limited to only hands.

Cons:

- 7500\$ price tag plus monthly subscriptions limit purchases to only big corporations⁸
- The corded solution limits the mobility of the user, limiting realism
- Premade hand interfaces limit potential use cases

⁷ Fundamental Surgery. (2024). HapticVR®. Fundamental Surgery. Retrieved 12 06, 2024, from <https://fundamentalsurgery.com/platform/hapticvr>

⁸ Slachta, A. (2018, 11 1). *FundamentalVR to develop novel virtual reality solutions*. HealthExec. Retrieved 12 06, 2024, from <https://healthexec.com/topics/patient-care/digital-transformation/mayo-clinic-partners-fundamentalvr>



Figure 8.1: An image showing the product in usage. The two objects on the left side of the screen control and record the trainee's motion. The light blue object in the middle is a detachable object used to replicate surgical tools.⁹



Figure 8.2: An image depicting the system in usage with a trainee operating the machine and a trainer guiding them through the procedure.¹⁰

⁹ Fundamental Surgery. (2024). *HapticVR®*. Fundamental Surgery. Retrieved 12 06, 2024, from <https://fundamentalsurgery.com/platform/hapticvr/#what-is-hapticvr>

¹⁰ Fundamental Surgery. (2024). *HapticVR®*. Fundamental Surgery. Retrieved 12 06, 2024, from <https://fundamentalsurgery.com/platform/hapticvr/#what-is-hapticvr>

Product 3: HaptX Gloves G1

Made by HaptX, this project simulates realistic touch for training, robotics, and other digital environments. This touch is created through microfluidic tactile actuators, small bags that inflate to physically move your skin to give specificity to the type of touch felt. Both gloves are attached to an Airpack, and strapped onto a user's back, delivering airflow to all actuators.¹¹

Specifications:

- Flexible force feedback mechanisms
 - 8 lbs per finger, 5 lb expected operating force
- 36 degrees of freedom
- Comes in four sizes: S, M, L, XL
- 16 ms latency, 120 Hz update rate
- Resolution: Positional: .01 in, Rotational: .3 deg
- Customizable buttons, knobs, switches, etc.
- HaptX SDK to implement gloves in supported software

Pros:

- Mocap functionality
- Developer tools and plugins for Unreal Engine and Unity integration
- Can discern textures through displacement maps
- Able to be improved with development visualizers
- Has both mobile and desktop uses

Cons:

- 20 lbs airpack may weigh over time
- Unclear configuration requirements
 - Wi-fi compatibility in testing
- Expensive
 - \$5,495 for an individual pair, \$18,000 for a bundle of all 4 sizes¹²
- Glove life expectancy of only 540 hours
 - Extensive medical use from multiple users would wear out in a few months
- Typical battery life of 2 hours
- No specific medical-based software
 - Required to create own simulations using SDK: technical knowledge required
- No onboard computing

¹¹ HaptX. (2024). *HaptX Gloves G1*. HaptX. <https://haptx.com/gloves-g1/>

¹² Keiretsu Forum. (2022, October 25). *HaptX introduces industry's most advanced haptic gloves, priced for scalable deployment*. <https://www.k4northwest.com/articles/haptx-introduces-industrys-most-advanced-haptic-gloves-priced-for-scalable-deployment>



Figure 9.1: Photo of the entire HaptX Gloves G1 system: both force-touch feedback gloves attached to an Airpack.¹³

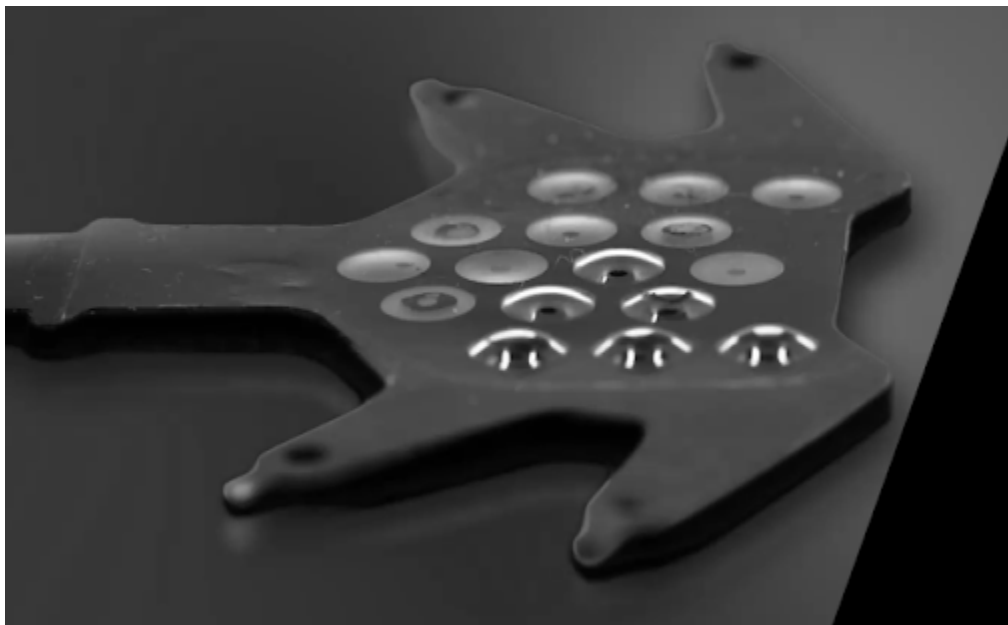


Figure 9.1: Photo of a microfluidic actuator panel. Hundreds of actuators line the glove's surface.¹⁴

¹³ Hayden, S. (2022, October 25). *HaptX's Latest Haptic Gloves Are Smaller and Cheaper, But Still Bulky and Expensive*. Road to VR. Retrieved December 8, 2024, from <https://www.roadtovr.com/haptx-haptic-gloves-g1-price-release-date/>

¹⁴ HaptX. (2024). *HaptX Gloves G1*. HaptX. <https://haptx.com/gloves-g1/>

Product 4: Anatomage Table

Made by Anatomage, their Table is a platform that offers digitized human cadavers and real patient cases as simulations on a huge tablet surface. Various scans of male, female, geriatric, and pregnant bodies can be viewed with high visualization scans, rendering bone, muscle, and tissue on their huge annotatable digital screen.¹⁵

Specifications:

- Various versions: Classic, Convertible, Clinical, Vet
 - Focus is on the Classic version
- 87" x 33" x 28" multi-touch screen with tempered glass overlay / Weight: 400 lbs
- Can be connected to Internet and external sources (projectors, monitors)
- Uses DICOM rendering to visualize scans
 - 3D models made from 2D slices
 - Able to render to .5mm of detail
- Abundant clinical database: 1,100+ digital histology scans, 77 dissected scans
 - Can simulate physical behaviors within scans

Pros:

- Huge clinical database with simulation allows for in-depth analysis of cadavers
- Multi-touch screen allows for collaboration
- FDA-certified information, providing a great educational experience
- Upgradable software

Cons:

- Extremely expensive: ~\$72,000¹⁶
- No physical hands-on experience: surgical training is limited
- Physically huge
 - Requires a lot of area for the product itself and surrounding users
- No functionality with Linux, OSX, Android, or iOS¹⁷
- Learning curve to use software

¹⁵ Anatomage. (n.d.). Anatomage Table | Digital Anatomy and Physiology Medical Learning Platform. Anatomage. Retrieved December 8, 2024, from <https://anatomage.com/table/>

¹⁶ Michigan Tech. (2022, September 1). Anatomage Table 8.0 is a True Game Changer for Students. Michigan Tech Biological Sciences Blog. <https://blogs.mtu.edu/biological/2022/09/anatomage-table/>

¹⁷ Anatomage, Inc. (2023, August 18). Anatomage Table EDU 10.0 User's Manual. Anatomage Table EDU 10.0 Application Manual. https://hsasf.hsa.washington.edu/wp-content/uploads/2023/12/Table_EDU_10.0_Application_User_s_Manual__Rev_A_.pdf



Figure 10.1: A photo of an Anatomage Table Classic.¹⁸



Figure 10.2: A photo of three medical professionals interacting with an Anatomage Table Classic.¹⁹

¹⁸ Anatomage. (n.d.). Anatomage Table | Digital Anatomy and Physiology Medical Learning Platform. Anatomage. Retrieved December 8, 2024, from <https://anatomage.com/table/>

¹⁹ Anatomage. (n.d.). Anatomage Table | Digital Anatomy and Physiology Medical Learning Platform. Anatomage. Retrieved December 8, 2024, from <https://anatomage.com/table/>

Product 5: Fluid Reality Haptic VR Glove

Created by the Future Interfaces Group, this untethered, lightweight, and cost-efficient haptic glove uses 160 haptic feedback actuators to deliver pressure and patterns to fingertips.²⁰ The glove uses high-resolution fingerpad arrays to imprint tactile images onto the skin, which can display different surfaces, edges, and even environmental effects like wind, water, and vibrations.²¹

Specifications:

- 160 dynamic haptic feedback actuators
- Weights around 0.2 kg
- Less than \$1000
- Untethered glove battery
- Uses an electrically controlled pump to cause electroosmosis
- Uses silicone membrane, PSA, PCB, and pump membrane

Pros:

- Lightweight, cheap, and wireless
- Uses off-the-shelf components like Raspberry Pi, so can be easily repaired

Cons:

- Hard to put on and take off with separated finger pads and multiple dangling parts
- Battery life may not be very long

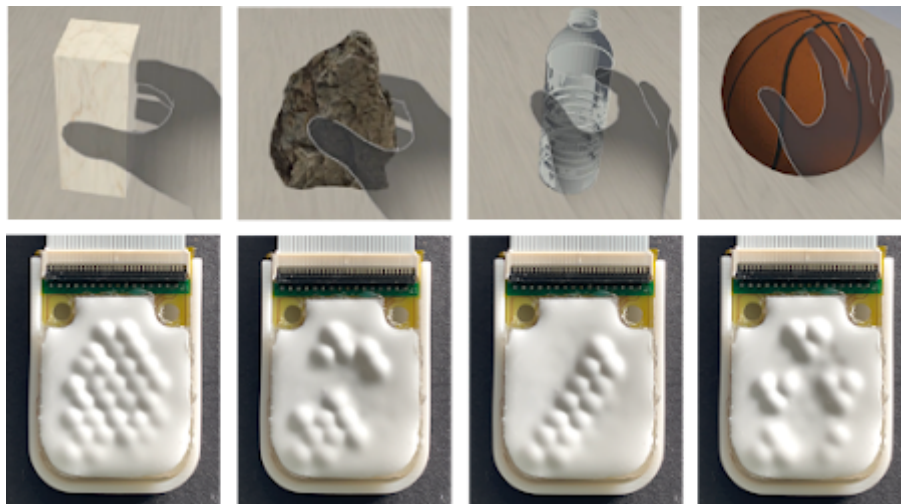


Figure 11.1: Different objects resulting in different tactile responses on the shape arrays.²²

²⁰ Reality, Fluid. "Fluid Reality Introduces Its Untethered, High-Res Haptic VR Gloves." *New Atlas*, 31 Oct. 2023, newatlas.com/vr/fluid-reality-introduces-its-untethered-high-res-haptic-vr-gloves/. Accessed 08 Dec. 2024.

²¹ "The Future of Touch." *Fluid Reality*, www.fluidreality.com/. Accessed 08 Dec. 2024.

²² "Fluid Reality." About, www.figlab.com/research/2023/FluidReality. Accessed 08 Dec. 2024.



Figure 11.2: Photo of fluid reality glove with all components attached to the hand and fingers.²³

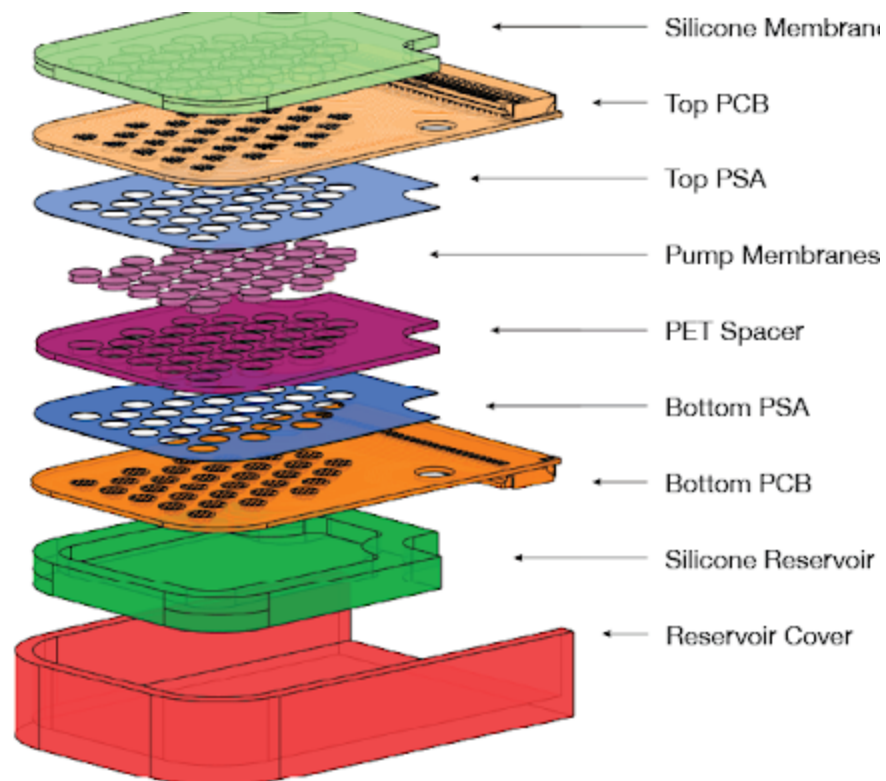


Figure 11.3: Exploded view of one high-res fingertip shape array.²⁴

²³ "The Future of Touch." *Fluid Reality*, www.fluidreality.com/. Accessed 08 Dec. 2024.

²⁴ "Fluid Reality." *About*, www.figlab.com/research/2023/FluidReality. Accessed 08 Dec. 2024.

Product 6: TouchDIVER Haptic Gloves

This haptic glove comes with multiple features such as force feedback and thermal cues, as well as precise sensors to track hand and finger movements.²⁵ From professional training to virtual prototyping to robotics, this haptic glove can work with any software to create immersive, tactile surgical simulations. TouchDIVER gloves can be accessed from anywhere as well, saving time and money.

Specifications:

- Multimodal haptic feedback
- 6 actuation points
 - Finger tracking accuracy of 2mm
 - Precision of 0.6 degrees²⁶
- 200 grams (~0.5 lbs)
- Thermal range between 18°C to 42°C (64.4°F to 107.6°F)
- One size fits all, flexible and stretchable materials
- Easy-to-use SDK for realistic and reliable manipulation
 - Supports Unity, Unreal Engine, C++/C#/Python²⁷
 - Compatible with Meta Quest, Pico, HTC, Windows Mixed Reality
 - Low latency with WiFi, Bluetooth, or tethered setups

Pros:

- Wearable and lightweight
- Physical, virtual, and thermal feedback
- Added palm actuation point
- Hot, swappable battery (fast battery replacements)
- Easy to put on and take off
- Can be connected to headsets or PCs easily
- Texture rendering

²⁵ “VR Medical Training: Weart.” *WEART Haptics*, 2 Dec. 2024, weart.it/use-cases/vr-medical-training/. Accessed 08 Dec. 2024.

²⁶ Kolo, Kris. “WEART Introduces the New TouchDIVER Pro Haptic Glove: A New Dimension for XR Interactions.” *VR/AR Association (VRARA)*, 12 June 2024, www.thevrara.com/blog2/2024/6/12/weart-introduces-the-new-touchdiver-pro-haptic-glove-a-new-dimension-for-xr-interactions. Accessed 08 Dec. 2024.

²⁷ “Weart TouchDIVER Pro Haptic Gloves.” *Knoxlabs*, www.knoxlabs.com/products/weart-touchdiver-pro-haptic-gloves?_pos=1&_sid=51b4e4463&_ss=r. Accessed 08 Dec. 2024.

Cons:

- Expensive: costs \$4900 for gloves
- Requires initial setup from VR headsets
- Complex and may have limited applications due to its recency



Figure 12.1: View of the TouchDIVER haptic gloves by itself (right) and when worn (left).²⁸



Figure 12.2: Sample illustration of using the VR glove in a potential medical training setting.²⁹

²⁸ “Weart TouchDIVER Pro Haptic Gloves.” Knoxlabs, www.knoxlabs.com/products/weart-touchdiver-pro-haptic-gloves?_pos=1&_sid=51b4e4463&_ss=r. Accessed 08 Dec. 2024.

²⁹ “VR Medical Training: Weart.” WEART Haptics, 2 Dec. 2024, weart.it/use-cases/vr-medical-training/. Accessed 08 Dec. 2024.

III. Product Comparison

SIMILAR SOLUTIONS MATRIX			1: Unfettered	2: Resists hand motion	3: Resists Arm Motion	4: Surface Texture Mimicing	5: Lightweight	6: Medical Field Specific	7: Multiple hand sizes	8: Usage Duration	9: Quick Reuse Time	10: Cheaper than 1000\$
Type	Sources	Solutions & Designs	Yes				Somewhat / Unknown				No	
Products	knoxlabs	bHaptics TactGlove DK1										
	hapticvr	HapticVR										
	haptx	HaptX Gloves G1										
	fluidreality	Fluid Reality Haptic VR Glove										
	knoxlabs2	TouchDIVER Haptic Gloves										
Patents	patentimage1	Exo-Tendon Haptic Glove (Patent 2018/0311570)										
	patentimage2	VR Jamming Garment/ Glove (Patent 10,976,826)										
	patentimage3	Pneumatic Augmented Reality Tactile Feedback Platform (Patent 2017/0300115)										

Figure 13.1: A similar solution matrix for 8 products/patents, showing their characteristics regarding 10 criteria.

Within the medical training field, there are a lot of solutions to improve hands-on training for medical staff. Each solution has unique pros and cons, making comparisons difficult, but a similar solution matrix can simplify the task. Products that succeed in an area/have a characteristic of interest are given a green box, whereas a red box signifies those that do not. If a product mostly does a task, or its functionality can not be determined, it is given a yellow box. Out of the eight solutions shown in Figure 13.1, TouchDIVER Haptic Gloves outperform the others. The ability to resist hand/finger motion, and support for multiple hand sizes, seem to be important characteristics as most solutions support them in one way or another.

IV. Conclusion

These solutions – products and patents – delineate the various forms of VR touch technology, medical instruments, or an intersection of both fields. The similar solution matrix concludes two key factors regarding methods to solve the problem: immersion, and authenticity.

All products focus on one factor but fail to deliver on the other. Various haptic feedback gloves simulate different versions of touch: sensation, texture, and immobility. However, these gloves are primarily made for general purposes, lacking specificity for medical simulations, and not authentic to an educational setting hoping to reduce cadaver use.

Likewise, simulations of medical processes in and out of the VR space lack the immersion of being in a situation similar to future work with patients or real cadavers, the main problem being the lack of touch.

However, the similar solution matrix allows for the first steps in understanding key features required in the final product, which should be tailored to medical processes, and simulate all touch features. This is the focus that the next steps will take to ensure an effective solution, through criteria, constraints, and design specifications.